Toric varieties

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Lecture 1

In this course, we do everything over $\mathbb{C}$.

1 What is a toric variety?

Definition 1.1 (affine variety)
An *affine variety* is $V = \mathrm{Spec}(R)$, where R is a finitely generated integral[a] $\mathbb{C}$-algebra.

[a] R is an integral domain.

Fact: Given such an R, we can choose generators and relations, and write

$$R = \frac{\mathbb{C}[x_1, \ldots, x_n]}{I}$$

where I is a prime ideal. This gives an affine embedding. i.e. a closed embedding

$$V = \mathbb{V}(I) = \mathbb{A}^n$$

Note that the affine embedding is not unique.

Definition 1.2 ((complex) variety)
A *(complex) variety* is a scheme X which is integral (reduced and irreducible), separated, and of finite type.

We can avoid schemes, and glue varieties out of affine varieties. We need

1. a finite collection $\{V_\alpha\}_{\alpha \in A}$ of affine varieties, say $V_\alpha = \mathrm{Spec}(R_\alpha)$,
2. for $\alpha, \beta \in A$, we need Zariski opens $V_{\alpha\beta} \subseteq V_\alpha$,
3. algebraic transition maps $g_{\alpha\beta} : V_{\alpha\beta} \to V_{\beta\alpha}$, satisfying the usual cocycle conditions,

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4. check that the resulting

$$X = \frac{\bigsqcup_\alpha V_\alpha}{\sim}$$

is irreducible and separated (Hausdorff).

Definition 1.3 (torus)
An *(algebraic complex) torus of dimension n* is

$$\mathrm{T}^n = (\mathbb{C}^*)^n = \mathrm{Spec}(\mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]) = (\mathbb{A}^1 \setminus \{0\})^n$$

Note that this is not the same as the usual torus $(S^1)^n$. In particular, we have an affine embedding

$$\mathrm{T}^n = \mathrm{Spec}\left(\frac{\mathbb{C}[z_0, \dots, z_n]}{\langle z_0 \cdots z_n - 1\rangle}\right) = \mathbb{V}(z_0 \cdots z_n - 1) \subseteq \mathbb{A}^{n+1}$$

Exercise: Check that we have an isomorphism of rings

$$\mathbb{C}[z_1^{\pm 1}, \dots, z_n^{\pm 1}] \cong \frac{\mathbb{C}[z_0, \dots, z_n]}{\langle z_0 \cdots z_n - 1\rangle}$$

The torus is also a *commutative group scheme*, that is, we have algebraic morphisms

$$\begin{aligned} e &: \bullet \to \mathrm{T}^n \\ m &: \mathrm{T}^n \times \mathrm{T}^n \to \mathrm{T}^n \\ i &: \mathrm{T}^n \to \mathrm{T}^n \end{aligned}$$

making T^n into an abelian group with identity e, multiplication m and inverse i. On the level of coordinate rings, we have the *identity section*,

$$\begin{aligned} e^\sharp : \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}] &\to \mathbb{C} \\ t_i^{\pm 1} &\mapsto 1 \end{aligned}$$

The multiplication map is given by pointwise operations, i.e.

$$m((x_1, \dots, x_n), (y_1, \dots, y_n)) = (x_1 y_1, \dots, x_n y_n)$$

With this, we have *comultiplication*

$$\begin{aligned} m^\sharp : \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}] &\to \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}] \otimes \mathbb{C}[y_1^{\pm 1}, \dots, y_n^{\pm 1}] \\ t_i &\mapsto x_i \otimes y_i \end{aligned}$$

and the inverse map is given by

$$\begin{aligned} i^\sharp : \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}] &\to \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}] \\ t_i &\mapsto t_i^{-1} \end{aligned}$$

Definition 1.4 (toric variety)
A *toric variety* is a variety X with a dense open $\mathrm{T}^n \subseteq X$, and an action (i.e. algebraic map)

$$a : \mathrm{T}^n \times X \to X$$

which extends the multiplication on X.

Example 1.5

T^n is itself a toric variety.

Cohomology

Example 1.6

$\mathbb{A}^n = \mathrm{Spec}(\mathbb{C}[x_1, \ldots, x_n])$, with the torus

$$\mathrm{Spec}(\mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}])$$

with

$$a((t_1, \ldots, t_n), (x_1, \ldots, x_n)) = (t_1 x_1, \ldots, t_n x_n)$$

Example 1.7

Projective space

$$\mathbb{P}^n = \mathrm{Proj}(\mathbb{C}[z_0, \ldots, z_n]) = \{(z_0 : \cdots : z_n) \mid (z_0, \ldots, z_n) \neq (0, \ldots, 0)\}$$

is also a toric variety, with torus

$$\mathrm{T}^n = \{(z_0 : \cdots : z_n) \mid z_0 \cdots z_n \neq 0\}$$

with action given coordinate-wise again, i.e.

$$a((t_0 : \cdots : t_n), (z_0 : \cdots : z_n)) = (t_0 z_0 : \cdots : t_n z_n)$$

in non-homogeneous coordinates (i.e. in the affine $z_0 \neq 0$) of the torus, we can define the action by

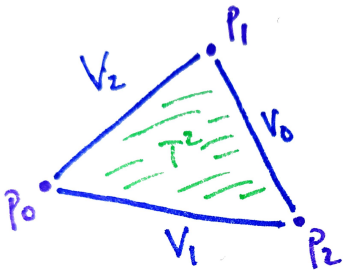
$$(t_1, \ldots, t_n) \cdot (z_0 : \cdots : z_n) = (z_0 : t_1 z_1 : \cdots : t_n z_n)$$

We can check that T^n is contained in all of the standard opens $U_i = D(z_i)$, and if we choose the usual affine coordinates, then T^n embeds into affine space as above.

Toric varieties are determined by the dimension n of the torus, and the *toric boundary* $X \setminus \mathrm{T}^n$. For example, $\mathbb{P}^2 \setminus \mathrm{T}^2$ has three components,

1. $V_0 = \mathbb{V}(z_0)$,
2. $V_1 = \mathbb{V}(z_1)$,
3. $V_2 = \mathbb{V}(z_2)$

The V_i intersect pairwise, but not all three at the same time. Each V_i is invariant under the action of the torus, so $\mathbb{P}^2$ can be represented as



In particular, this gives us a stratification of $\mathbb{P}^2$, with strata T^2, $\mathbb{C}^* = T^1$ corresponding to V_i with the vertices removed, and the three vertices.

2 Characters, cocharacters and lattices

Recall from the last lecture that $\mathrm{T}^n = (\mathbb{C}^*)^n$ is an algebraic group.

Definition 2.1 (character)
A *character* χ of T^n is a morphism $\chi : \mathrm{T}^n \to \mathbb{C}^*$ of algebraic groups. That is, $\chi : \mathrm{T}^n \to \mathbb{C}^*$ is a morphisms of schemes, such that

$$\chi(e) = 1 \quad \text{and} \quad (\chi^\sharp \otimes \chi^\sharp) \circ m^\sharp = m^\sharp \circ \chi^\sharp$$

We will write $\mathrm{Hom}_{\text{alg-gp}}(\cdot, \cdot)$ for the set of morphisms of algebraic groups.

Theorem 2.2. The characters of T^n are

$$M = \mathrm{Hom}_{\text{alg-gp}}(\mathrm{T}^n, \mathbb{C}^*) = \mathbb{Z}^n$$

Proof. First of all, note that

$$\mathrm{Hom}_{\text{alg-gp}}(\mathrm{T}^n, \mathbb{C}^*) = (\mathrm{Hom}(\mathbb{C}^*, \mathbb{C}^*))^n$$

Thus, without loss of generality $n = 1$. Now we need a $\mathbb{C}$-algebra map

$$\chi^\sharp : \mathbb{C}[x^{\pm 1}] \to \mathbb{C}[y^{\pm 1}]$$

This is determined by $\chi^\sharp(x) = p(y, y^{-1})$. Note that x is a unit in $\mathbb{C}[x^{\pm 1}]$, and so $\chi^\sharp(x)$ is a unit. Thus, $p(y, y^{-1}) = ay^k$, for some $a \in \mathbb{C}, k \in \mathbb{Z}$.

Next, we need that $e^\sharp \circ \chi^\sharp = \chi^\sharp \circ e^\sharp$. That is,

$$a = e^\sharp(\chi^\sharp(x)) = \chi^\sharp(e^\sharp(x)) = \chi^\sharp(1) = 1$$

Hence $\chi^\sharp(x) = x^k$, for some $k \in \mathbb{Z}$. Note that $\chi^\sharp(x \otimes x) = \chi^\sharp(x) \otimes \chi^\sharp(x)$ automatically, so we are done. □

The characters of T^n are of the form

$$\chi(t_1, \ldots, t_n) = t_1^{a_1} \cdots t_n^{a_n}$$

for some $(a_1, \ldots, a_n) \in \mathbb{Z}^n$. Expressions of the form

$$t_1^{a_1} \cdots t_n^{a_n}$$

are called *Laurent monomials*. We will write

$$\chi^{\mathbf{a}}(\mathbf{t}) = \mathbf{t}^{\mathbf{a}} = t_1^{a_1} \cdots t_n^{a_n}$$

for this character. The correspondence is then

$$\begin{aligned} \mathrm{Hom}_{\text{alg-gp}}(\mathrm{T}^n, \mathbb{C}^*) &\to \mathbb{Z}^n \\ \chi^{\mathbf{a}} &\mapsto \mathbf{a} \end{aligned}$$

The characters form a group. Given characters $\chi, \chi' : \mathrm{T}^n \to \mathbb{C}^*$,

$$\chi \cdot \chi'(\mathbf{t}) = \chi(\mathbf{t}) \cdot \chi'(\mathbf{t})$$

Thus, we have taht

$$\chi^{\mathbf{a}} \cdot \chi^{\mathbf{a}'} = \chi^{\mathbf{a}+\mathbf{a}'}$$

Hence the identification $\mathrm{Hom}_{\text{alg-gp}}(\mathrm{T}^n, \mathbb{C}^*) \cong \mathbb{Z}^n$ is also a group isomorphism.

Definition 2.3 (cochacter)
A *cocharacter* of T^n is

$$\lambda \in \mathrm{Hom}_{\text{alg-gp}}(\mathbb{C}^*, \mathsf{T}^n)$$

We also call them *one-parameter subgroup*.

We can check that

$$N = \mathrm{Hom}_{\text{alg-gp}}(\mathbb{C}^*, \mathsf{T}^n) \cong \mathbb{Z}^n$$

by showing that any $\lambda \in N$ is of the form

$$\lambda(t) = (t^{u_1}, \dots, t^{u_n})$$

for $\mathbf{u} = (u_1, \dots, u_n) \in \mathbb{Z}$. So we can write the above cocharacter as $\lambda_{\mathbf{u}}$ as before. N is a group, and the identification is a group isomorphism. That is,

$$\lambda_{\mathbf{u}} \cdot \lambda_{\mathbf{u}'} = \lambda_{\mathbf{u}+\mathbf{u}'}$$

Example 2.4
For $n = 2$, we can identify

$$\mathsf{T}^2 = \left\{ \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} \;\middle|\; t_1 t_2 \neq 0 \right\}$$

Under the identification $M \cong N \cong \mathbb{Z}^2$,

$$\chi^{(a,b)} \begin{pmatrix} t_1 & 0 \\ 0 & t_2 \end{pmatrix} = t_1^a \cdot t_2^b$$

and

$$\lambda_{(u,v)}(t) = \begin{pmatrix} t^u & 0 \\ 0 & t^v \end{pmatrix}$$

Definition 2.5 (lattice)
A *lattice* is a free abelian group of finite rank.

Note that with this definition, any lattice is isomorphic to $\mathbb{Z}^n$ when we choose a basis.

Example 2.6
$\mathbb{Z}, 2\mathbb{Z}, \mathbb{Z} \oplus \mathbb{Z}, \mathbb{Z} \oplus 3\mathbb{Z}, \dots$ are all lattices.

There is a *perfect pairing*[1], given by

$$\begin{aligned} \langle \cdot, \cdot \rangle : M \times N &\to \mathbb{Z} \\ \langle \mathbf{a}, \mathbf{u} \rangle = \mathbf{a} \cdot \mathbf{u} &= a_1 u_1 + \cdots + a_n u_n \end{aligned}$$

In a coordinate-free manner,

$$\begin{aligned} M \times N = \mathrm{Hom}(\mathbb{C}^*, \mathsf{T}^n) \times \mathrm{Hom}(\mathsf{T}^n, \mathbb{C}^*) &\to \mathrm{Hom}(\mathbb{C}^*, \mathbb{C}^*) \\ (\lambda, \chi) &\mapsto \chi \circ \lambda \end{aligned}$$

Using the identification $\mathrm{Hom}(\mathbb{C}^*, \mathbb{C}^*) \cong \mathbb{Z}$ as before gives the pairing.

We can describe a torus T_N starting from a lattice N, through

$$\mathsf{T}_N = N \otimes_{\mathbb{Z}} \mathbb{C}^*$$

[1] That is, it induces isomorphism $\mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Z}) = M^\vee \cong N$ and vice versa.

with this,

$$\mathrm{Hom}(\mathbb{C}^*, \mathrm{T}_N) = N$$

Formally,

$$\mathrm{T}_N = \mathrm{Spec}(\mathbb{Z}[N] \otimes_{\mathbb{Z}} \mathbb{C}[x, x^{-1}])$$

With this, give a torus T, we get two dual lattices N, M. Conversely, given a lattice N, we get a torus T_N and a dual lattice M. Choosing a basis on one gives a basis on the other two.

3 Cocharacters and limit points - Part I

Back to toric varieties. Let X be a toric variety, with torus T. We can think of X as a "partial compactification" of T. That is, we can construct X by adding limit points[2] to T.

Given a one-parameter subgroup $\mathbf{u} \in N$, we have

$$\lambda_{\mathbf{u}} : \mathbb{C}^* \to \mathrm{T} \subseteq X$$

We can ask when does

$$\lim_{t \to 0} \lambda_{\mathbf{u}}(t)$$

exist in X. The idea will be that we can determine X by saying which $\lambda_{\mathbf{u}}$ have a limit. See later.

Example 3.1

Consider $\mathrm{T}^2 \subseteq \mathbb{P}^2_{z_0:z_1:z_2}$. The embedding is given by $(t_1, t_2) \mapsto (1 : t_1 : t_2)$. Choose $\mathbf{u} \in \mathbb{Z}^2$, so $\mathbf{u} = (u_1, u_2)$.

$$\lim_{t \to 0} \lambda_{\mathbf{u}}(t) = \lim_{t \to 0} (1 : t^{u_1} : t^{u_2})$$

All these limit points exist, since $\mathbb{P}^2$ is complete[a]. In particular, if $u_1, u_2 > 0$, the limit is $(1 : 0 : 0)$. If $u_1 < 0, u_2 > u_1$, the limit is $(0 : 1 : 0)$. Similarly if $u_2 < 0, u_1 > u_2$ then the limit is $(0 : 0 : 1)$.

The upshot is that we have partitioned $N_{\mathbb{R}} = N \otimes \mathbb{R} = \mathbb{R}^2$ into cones, so that if $\mathbf{u}_1, \mathbf{u}_2$ are in the same cone if and only if they have the same limit. In fact, we can construct $\mathbb{P}^2$ from this.

[a] Or in algebraic geometry, it satisfies the valuative criterion for properness for some choice of DVR.

Lecture 3

4 Affine toric varieties and cones

Definition 4.1 (cone)

A *(rational polyhedral) cone* is

$$\sigma \subseteq N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R} = \mathbb{R}^n$$

where

$$\sigma = \sum_{i \in I} \lambda_i u_i \mid \lambda_i \geq 0, u_1, \ldots, u_n \in \mathbb{N}$$

We will also write

$$\sigma = \mathrm{Cone}(u_1, \ldots, u_r)$$

and the set $\{u_i\}_{i=1}^r$ is the set of generators of the cone.

Example 4.2

If $N_{\mathbb{R}} \cong \mathbb{R}e$, then we have the cone $\sigma_1 = \mathrm{Cone}(e)$,
Next, $\sigma_2 = \mathrm{Cone}((1, 0))$ is
and $\sigma_3 = \mathrm{Cone}((1, 0), (0, 1))$ is

[2] In the Euclidean topology.

and $\sigma_4 = \text{Cone}((1,0),(1,2))$ is

Definition 4.3 (dimension)
The *dimension* of a cone σ is $\dim(\sigma) = \dim(V)$, where $V = \text{span}\{\sigma\} \subseteq N_{\mathbb{R}}$. A cone is *top-dimensional* if $V = N_{\mathbb{R}}$.

Definition 4.4 (strongly convex)
σ is *strongly convex* if it does not contain any lines through the origin. Equivalently, $\sigma \cap -\sigma = \{0\}$.

We would like to associate a to a cone σ, an affine toric variety $U_\sigma = \text{Spec}(R_\sigma)$. That is, we would like to define

$$R_d \subseteq \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$$

a sub-$\mathbb{C}$-algebra.

We will do this as follows:

1. We will associate a *dual cone* $\sigma^\vee \subseteq M_{\mathbb{R}}$,
2. Then we will generate a commutative semigroup S_σ,
3. from this, we will define R_σ.

4.1 Dual cone

Definition 4.5 (dual cone)
The *dual cone* $\sigma^\vee$ of σ is

$$\check{\sigma} = \{m \in M_{\mathbb{R}} \mid \langle u, m\rangle \geq 0 \text{ for all } u \in \sigma\} \subseteq M_{\mathbb{R}}$$

where we extend the pairing to $N_{\mathbb{R}} \otimes M_{\mathbb{R}} \to \mathbb{R}$.

Example 4.6
Using the same examples as before, we have the dual cones $\check{\sigma_1}$
and $\check{\sigma_2}$,
Note here we see that the dual cone depends on the embedding (or lattice). Moroever, the dual of a strongly convex cone does not have to be strongly convex.

Fact: $\check{\check{\sigma}} = \sigma$.

Lemma 4.7. The dual $\check{\sigma}$ of a cone σ is a cone.

Proof. We'll define an algorithm to find $m_1, \dots, m_s \in M$, such that

$$\check{\sigma} = \text{Cone}(m_1, \dots, m_s)$$

From linear algebra, for $m \in M = N^\vee$, we have a hypersurface

$$m^\perp = H_m = \{u \in N_{\mathbb{R}} \mid \langle u, m\rangle = 0\}$$

We also have half spaces

$$H_m^+ = \{u \in N_{\mathbb{R}} \mid \langle m, u\rangle \geq 0\}$$

Definition 4.8 (supporting hyperplane)
A *supporting hyperplane* of σ is H_m, such that $\sigma \subseteq H_m^+$. Equivalently, $m \in \check{\sigma}$.

Now note that

$$\check{\sigma} = \text{Cone}(m_1, \dots, m_s) \iff \sigma = \bigcap_{i=1}^{s} H_{m_i}^+$$

So to construct $\check{\sigma}$, we need to express σ as an intersection of half spaces.

For example, $\check{\sigma}_3 = \text{Cone}((1,0),(0,1))$

and similarly, $\check{\sigma}_4 = \text{Cone}((2,-1),(0,1))$.

In general, we can do this by identifying the *facets* $\tau \preccurlyeq \sigma$, and taking the corresponding dual element $m_\tau \in M$, and

$$\check{\sigma} = \text{Cone}(m_\tau \mid \tau \preccurlyeq \sigma)$$

when σ is top dimensional.

Definition 4.9
A *face* $\tau \preccurlyeq \sigma$ is the intersection $\sigma \cap H_m$, for a supporting hyperplane H_m. Note that the faces are all cones, and we call τ a *facet* if $\dim(\tau) = \dim(\sigma) - 1$, and τ an *edge* if $\dim(\tau) = 1$.

Note by taking $m = 0$, $\sigma \preccurlyeq \sigma$.

When σ is not top dimensional, then $\text{span}_{\mathbb{R}}(\sigma) = W \neq N_{\mathbb{R}}$. Write $\overline{\sigma}$ for σ considered as a subspace of W. Now $\overline{\sigma}$ is top-dimensional, and so we can the dual

$$\check{\overline{\sigma}} \subseteq \frac{M_{\mathbb{R}}}{W^\perp}$$

Say

$$\check{\overline{\sigma}} = \text{Cone}(\overline{m_1}, \dots, \overline{m_s})$$

where

$$\overline{m_i} \in \frac{M_{\mathbb{R}}}{W^\perp}$$

Choose representatives $m_i \in M_{\mathbb{R}}$, and choose a basis $m_{s+1}, \dots, m_k$ of $W^\perp$ in $M_{\mathbb{R}}$. With this,

$$\check{\sigma} = \text{Cone}(m_1, \dots, m_s, \pm m_{s+1}, \dots, \pm m_k)$$

□

4.2 Commutative semigroup

Definition 4.10 (commutative semigroup)
A *commutative semigroup* is $(S, +)$, such that $+$ is an associative commutative binary operation on S.

In particular, a semigroup with identity is called a *monoid*. S is *finitely generated* if there exists $A \subseteq S$ finite, such that

$$S = \left\{ \sum_n \lambda_n a_n \;\middle|\; \lambda_n \in \mathbb{Z}_{>0}, a_n \in A \right\}$$

S is *affine* if it is finitely generated, and $(S, +)$ is a subsemigroup of a lattice $(\mathbb{Z}^n, +)$.

Example 4.11
Any group is a semigroup. Given a (not necessarily unital) ring R, R under multiplication is a semigroup. For example, $\mathbb{Z}_{>0}$ is a semigroup under multiplication, but this is not finitely generated.

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